

Introduction

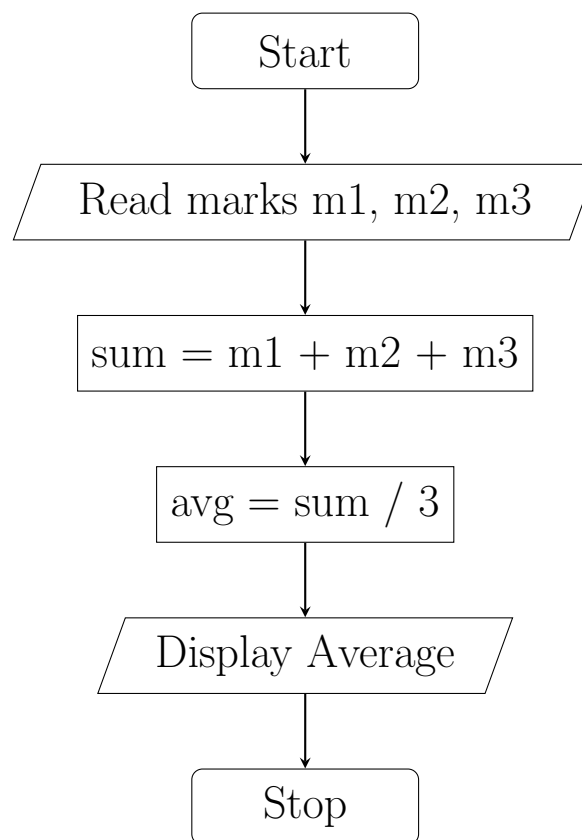
Python is a powerful, high-level, and interpreted programming language that emphasizes code readability and simplicity. It was developed by Guido van Rossum and officially released in 1991. Python was designed with the philosophy of making programming easy to learn and use, while still being capable of handling complex and large-scale applications. One of the main strengths of Python is its simple and clean syntax, which closely resembles the English language. This feature reduces the complexity of writing programs and makes it an ideal language for beginners. At the same time, Python is widely used by professionals for advanced development tasks. It supports multiple programming paradigms, including procedural programming, object-oriented programming (OOP), and functional programming, making it highly flexible and versatile. Python provides a vast standard library and a large ecosystem of third-party libraries and frameworks. These libraries allow developers to perform tasks such as web development, data analysis, machine learning, artificial intelligence, scientific computing, game development, automation, and cybersecurity. Frameworks like Django and Flask are used for web applications, while libraries such as NumPy, Pandas, and TensorFlow are widely used in data science and AI projects. Another important feature of Python is its platform independence. Programs written in Python can run on various operating systems such as Windows, macOS, and Linux without significant modifications. Python is also an open-source language, which means it is freely available and continuously improved by a large global community of developers. This assignment aims to explore the fundamental concepts of Python programming. It focuses on understanding variables, data types, operators, conditional statements, loops, functions, modules, file handling, and object-oriented programming concepts. Through practical implementation and problem-solving exercises, this assignment helps in developing logical thinking, analytical skills, and efficient coding practices. In conclusion, Python is a highly versatile and user-friendly programming language that plays a significant role in modern software development and technological innovation. Learning Python provides a strong foundation for future studies and career opportunities in various fields of computer science and information technology.

Problem Statement-1:-

Develop a Python program that reads marks from three different subjects and calculates their average.

❑ **Objective:-** The objective of this assignment is to develop a Python program that reads marks obtained in three different subjects and computes their average. Through this task, students will apply fundamental programming concepts such as input and output operations, variable declaration, data type handling, and arithmetic calculations. The assignment aims to strengthen understanding of basic syntax and program flow while enhancing problem-solving skills and logical reasoning in Python.

❑ **Flowchart:-**



❑ Algorithm:-

Algorithm 1 Average Calculation Program

```
1: Start
2: Declare  $m_1, m_2, m_3$  as real numbers
3: Declare  $sum, avg$  as real numbers
4: Read  $m_1, m_2, m_3$ 
5:  $sum \leftarrow m_1 + m_2 + m_3$ 
6:  $avg \leftarrow sum/3$ 
7: Display  $avg$ 
8: Stop
```

❑ Code:-

```
class Student:
    def __init__(student, name, *marks):
        student.name = name
        student.marks = list(marks)

    def avg(student):
        try:
            return sum(student.marks)/len(student.marks)
        except ZeroDivisionError:
            raise ValueError("At Least one value needed")

    def __repr__(student):
        return f"Student : {student.name}\nAverage : {student.avg()}"

s1 = Student(input("Enter Student name: "),
             int(input("Enter marks in maths: ")),
             int(input("Enter marks in Computer Science: ")),
             int(input("Enter marks in Electronics: ")))
print(s1)
```

❑ Output:-

```
C:\Users\DEBAJYOTI PARIA\OneDrive\Desktop>py student.py
Enter Student name: Debajyoti
Enter marks in maths: 50
Enter marks in Computer Science: 80
Enter marks in Electronics: 73
Student : Debajyoti
Average : 67.66666666666667

C:\Users\DEBAJYOTI PARIA\OneDrive\Desktop>|
```

Figure 1

- ❑ Conclusion:- In this assignment, a Python program was developed to read marks from three different subjects and calculate their average. The program demonstrates the practical use of basic Python concepts such as variables, input and output functions, and arithmetic operators.

By taking user input and applying the average formula:

$$\text{Average} = \frac{\text{Subject1} + \text{Subject2} + \text{Subject3}}{3}$$

the program shows how Python can be used to solve simple real-life problems efficiently and accurately.

This exercise improves logical thinking and helps in understanding how data is processed within a program. It also strengthens the foundation of programming concepts required for building more advanced applications in the future.

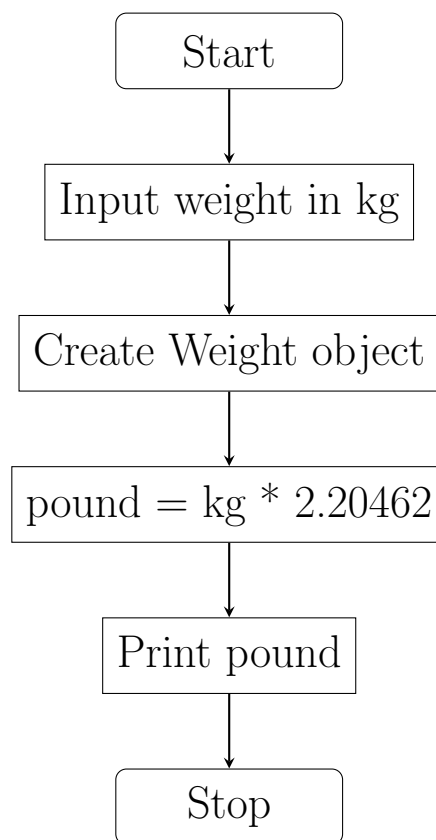
Overall, the objective of calculating the average marks was successfully achieved, highlighting the simplicity and effectiveness of Python programming.

Problem Statement-2:-

Create a Python script to convert weight from kilograms to pounds.

❑ **Objective:-** The objective of this assignment is to create a Python script that converts weight from kilograms to pounds using the standard conversion factor. This task helps in understanding user input handling, variable usage, arithmetic operations, and type conversion in Python. By implementing the formula (1 kilogram = 2.20462 pounds), students will practice writing simple computational programs and strengthen their grasp of basic programming logic and mathematical expressions.

❑ **Flowchart:-**



❑ Algorithm:-

Algorithm 2 Convert Kilogram to Pound Using Dataclass

- 1: Start
 - 2: Import dataclass from dataclasses module
 - 3: Define class Weight with attribute kg (float)
 - 4: Define property method pound
 - 5: Return $\text{kg} \times 2.20462$
 - 6: Read weight in kg from user
 - 7: Convert input to integer
 - 8: Create object w of class Weight using kg
 - 9: Print w.pound
 - 10: Stop
-

❑ Code:-

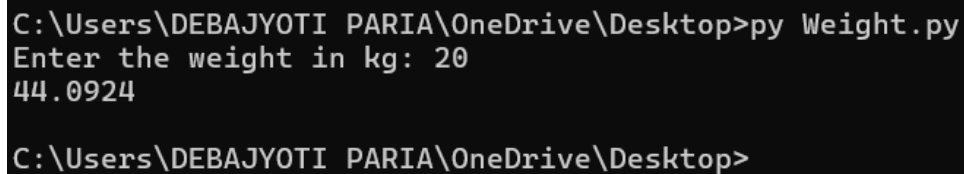
```
from dataclasses import dataclass

@dataclass
class Weight:
    kg: float

    @property
    def pound(self):
        return self.kg*2.20462

w = Weight(int(input("Enter the weight in kg: ")))
print(w.pound)
```

❑ Output:-



```
C:\Users\DEBAJYOTI PARIA\OneDrive\Desktop>py Weight.py
Enter the weight in kg: 20
44.0924

C:\Users\DEBAJYOTI PARIA\OneDrive\Desktop>
```

Figure 2

❑ **Conclusion:-** This program successfully demonstrates the use of Python's dataclass to create a simple and structured class for weight conversion. By defining a property method, the conversion from kilogram to pound is performed automatically without requiring an explicit method call.

The program takes user input, stores the value in an object, and calculates the equivalent weight in pounds using the standard conversion factor (1 kg = 2.20462 pounds). This approach improves code readability, modularity, and maintainability.

Overall, the implementation shows how object-oriented programming concepts such as classes, attributes, and properties can be effectively used to solve real-world problems in a clean and efficient manner.

Problem Statement-3:-

Write a program that takes the lengths of the three sides of a prism as input (integers) and computes its surface area using the formula: $2ab + 2bc + 2ca$.

❑ **Objective:-** The objective of this assignment is to develop a Python program that accepts the lengths of the three sides of a prism as integer inputs and calculates its surface area using the formula ($2ab + 2bc + 2ca$). This task reinforces understanding of user input processing, integer data types, arithmetic operations, and mathematical formula implementation in Python. It also helps students strengthen their logical reasoning and ability to translate mathematical expressions into executable code.

❑ Algorithm:-

Algorithm 3 Newton-Raphson Method

- 1: Choose initial guess x_0
 - 2: **repeat**
 - 3: $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$
 - 4: **until** $|x_{k+1} - x_k| < \epsilon$
 - 5: Return x_{k+1} as root
-

❑ Code:-

```
#include <stdio.h>
#include <math.h>

#define EPS 1e-6
#define MAX_ITER 100

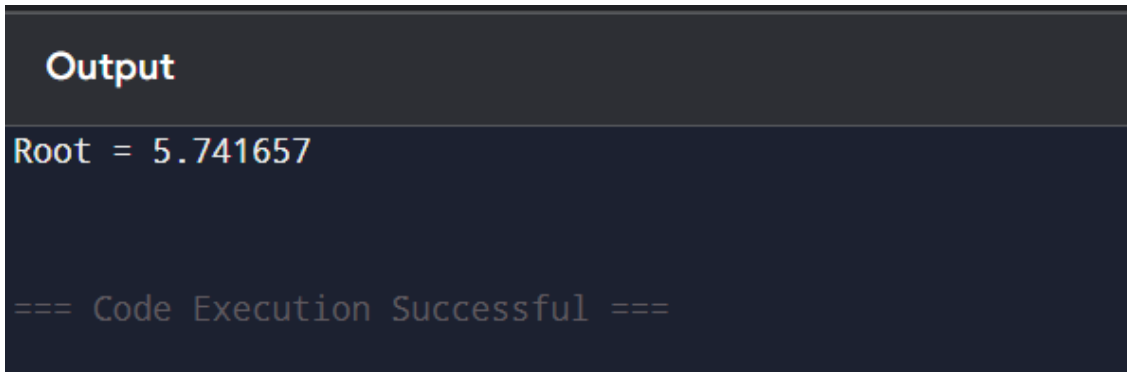
double f(double x) { return x*x - 4*x - 10; }
double df(double x) { return 2*x - 4; }

int main() {
```



```
double x0 = 5.0, x1;  
int iter = 0;  
  
do {  
    if (fabs(df(x0)) < EPS) {  
        printf("Derivative too small. Choose another initial  
        guess.\n");  
        return 1;  
    }  
    x1 = x0 - f(x0)/df(x0);  
    if (fabs(x1 - x0) < EPS) break;  
    x0 = x1;  
    iter++;  
} while(iter < MAX_ITER);  
  
printf("Root = %lf\n", x1);  
return 0;  
}
```

❑ Output:-



Output

Root = 5.741657

=== Code Execution Successful ===

Figure 3

❑ Conclusion:-

Problem Statement-4:-

Design a Python program to calculate the speed of a plane, given a distance of 395,000 meters and a travel time of 9,000 seconds.

❑ **Objective:-** The objective of this assignment is to design a Python program that calculates the speed of a plane using a given distance of 395,000 meters and a travel time of 9,000 seconds. This task helps students understand the practical application of the speed formula ($\text{Speed} = \frac{\text{Distance}}{\text{Time}}$). It reinforces concepts such as variable declaration.

❑ **Algorithm:-**

Algorithm 4 Gauss-Jordan Method for Matrix Inversion

- 1: Form augmented matrix $[A|I]$
 - 2: **for** $i = 1$ to n **do**
 - 3: Divide i -th row by pivot a_{ii} to make pivot = 1
 - 4: **for** $j = 1$ to $n, j \neq i$ **do**
 - 5: Subtract a_{ji} times i -th row from j -th row
 - 6: **end for**
 - 7: **end for**
 - 8: Resulting matrix is $[I|A^{-1}]$
-

❑ **Code:-**

```
#include <stdio.h>
#define N 3

int main() {
    float a[N][2*N], ratio;
    int i, j, k;

    printf("Enter matrix (%dx%d):\n", N, N);
    for(i = 0; i < N; i++)
        for(j = 0; j < N; j++)
            scanf("%f", &a[i][j]);
```

```

// Append identity matrix
for(i = 0; i < N; i++)
    for(j = N; j < 2*N; j++)
        a[i][j] = (i == (j-N)) ? 1 : 0;

// Gauss-Jordan
for(i = 0; i < N; i++) {
    float diag = a[i][i];
    for(j = 0; j < 2*N; j++) a[i][j] /= diag;
    for(k = 0; k < N; k++) {
        if(k != i) {
            ratio = a[k][i];
            for(j = 0; j < 2*N; j++)
                a[k][j] -= ratio * a[i][j];
        }
    }
}

printf("Inverse matrix:\n");
for(i = 0; i < N; i++) {
    for(j = N; j < 2*N; j++)
        printf("%8.4f ", a[i][j]);
    printf("\n");
}
return 0;
}

```

❑ Output:-

❑ Conclusion:-

```

Output
Enter matrix (3x3):
3 2 1
4 8 9
7 5 6
Inverse matrix:
0.0588 -0.1373 0.1961
0.7647 0.2157 -0.4510
-0.7059 -0.0196 0.3137

=== Code Execution Successful ===

```

Figure 4

Problem Statement-5:-

Build a program that determines how long it will take to empty a swimming pool of dimensions 12m $\times$ 7m $\times$ 2m using a pump with a flow rate of 17 cubic meters per hour.

❑ Objective:- The objective of this assignment is to develop a Python program that calculates the time required to empty a swimming pool based on its dimensions and the pump's flow rate. The program should first compute the volume of the pool using the formula ($\text{Volume} = \text{length} \times \text{width} \times \text{height}$), and then determine the time required to empty the pool using the formula ($\frac{\text{Volume}}{\text{Flow Rate}}$). This task strengthens understanding of arithmetic operations, world problem modeling in Python. It also enhances logical reasoning by

❑ Algorithm:-

Algorithm 5 Lagrange Interpolation

```

1: Given  $n$  data points  $(x_i, y_i)$ 
2: Initialize  $P(x) = 0$ 
3: for  $i = 1$  to  $n$  do
4:    $L_i(x) = \prod_{j=1, j \neq i}^n \frac{x - x_j}{x_i - x_j}$ 
5:    $P(x) \leftarrow P(x) + y_i L_i(x)$ 
6: end for
7: Return interpolated value  $P(x)$ 

```

□ Code:-

```

#include <stdio.h>

int main() {
    int n, i, j;
    float x[10], y[10], xp, yp=0, p;

    printf("Enter number of data points: ");
    scanf("%d", &n);
    printf("Enter x and y values:\n");
    for(i=0; i<n; i++) scanf("%f %f", &x[i], &y[i]);

    printf("Enter interpolation point: ");
    scanf("%f", &xp);

    for(i=0; i<n; i++) {
        p = 1;
        for(j=0; j<n; j++) {
            if(j != i) p *= (xp - x[j]) / (x[i] - x[j]);
        }
        yp += p*y[i];
    }
    printf("Interpolated value = %f\n", yp);
    return 0;
}

```

❑ Output:-

```
Enter number of data points:4
Enter x and y values:
1
1
2
2
3
3
4
4
Enter interpolation point:5
Interpolated value = 5.000000
```

Figure 5

❑ Conclusion:-

Problem Statement-6:-

Write a Python script to convert temperature from Celsius to Fahrenheit (input as a float).

- ❑ **Objective:-** The objective of this assignment is to develop a Python script that converts temperature from Celsius to Fahrenheit, where the Celsius value is provided as a float input. The program should implement the standard conversion formula ($F = \frac{9}{5}C + 32$). This task reinforces concepts such as user input handling, floating-point datatypes, arithmetic

□ Algorithm:-

(a) Trapezoidal Rule

Algorithm 6 Trapezoidal Rule

```

1: Divide interval  $[a, b]$  into  $n$  equal parts,  $h = \frac{b-a}{n}$ 
2:  $I = f(a) + f(b)$ 
3: for  $i = 1$  to  $n - 1$  do
4:    $I \leftarrow I + 2f(a + ih)$ 
5: end for
6:  $I \leftarrow \frac{h}{2} I$ 
7: Return  $I$ 

```

(b) Simpson's 1/3 Rule

Algorithm 7 Simpson's 1/3 Rule

```

1: Divide interval  $[a, b]$  into even  $n$  parts,  $h = \frac{b-a}{n}$ 
2:  $I = f(a) + f(b)$ 
3: for  $i = 1$  to  $n - 1$  do
4:   if  $i$  is odd then
5:      $I \leftarrow I + 4f(a + ih)$ 
6:   else
7:      $I \leftarrow I + 2f(a + ih)$ 
8:   end if
9: end for
10:  $I \leftarrow \frac{h}{3} I$ 
11: Return  $I$ 

```

□ Code:-

a) Trapezoidal Rule:

```

#include <stdio.h>
#include <math.h>

```

```

double f(double x) { return 1/(1+x*x); }

```

```

int main() {
    int n, i;
    double a, b, h, sum=0;

```



```

    printf("Enter a, b, n: ");
    scanf("%lf %lf %d", &a, &b, &n);
    h = (b-a)/n;
    sum = f(a)+f(b);
    for(i=1; i<n; i++) sum += 2*f(a+i*h);
    sum = sum*h/2;
    printf("Trapezoidal = %lf\n", sum);
    return 0;
}

```

b) Simpson's 1/3 Rule:

```

#include <stdio.h>
#include <math.h>

double f(double x) { return 1/(1+x*x); }

int main() {
    int n, i;
    double a, b, h, sum=0;
    printf("Enter a, b, n (even): ");
    scanf("%lf %lf %d", &a, &b, &n);
    h = (b-a)/n;
    sum = f(a)+f(b);
    for(i=1; i<n; i++)
        sum += (i%2==0 ? 2 : 4)*f(a+i*h);
    sum = sum*h/3;
    printf("Simpson 1/3 = %lf\n", sum);
    return 0;
}

```

❑ Output:-

```
Enter a, b, n:1
2
3
Trapezoidal = 0.324902
```

Figure 6.1

```
Enter a, b, n (even):1
2
3
Simpson 1/3 = 0.296601
```

Figure 6.2

❑ Conclusion:-

Problem Statement-7:-

Create a program to compute the total number of seconds in one full day.

❑ **Objective:-** The objective of this assignment is to develop a Python program that computes the total number of seconds in one full day. The program should apply the relationship between time units by converting hours to minutes and minutes to seconds using appropriate arithmetic operations. This task reinforces understanding of constant values, multiplication operations, and logical structuring of simple computational problems in Python. It also helps students strengthen their ability to translate real-world time calculations into precise program logic.

❑ **Algorithm:-**

(a) Euler's Method

Algorithm 8 Euler's Method

```

1: Input  $x_0, y_0, h, x_n$ 
2:  $x \leftarrow x_0, y \leftarrow y_0$ 
3: while  $x < x_n$  do
4:    $y \leftarrow y + hf(x, y)$ 
5:    $x \leftarrow x + h$ 
6: end while
7: Return  $y$ 

```

(b) Runge-Kutta 4th Order**Algorithm 9** Runge-Kutta 4th Order

```

1: Input  $x_0, y_0, h, xn$ 
2:  $x \leftarrow x_0, y \leftarrow y_0$ 
3: while  $x < xn$  do
4:    $k_1 = hf(x, y)$ 
5:    $k_2 = hf(x + h/2, y + k_1/2)$ 
6:    $k_3 = hf(x + h/2, y + k_2/2)$ 
7:    $k_4 = hf(x + h, y + k_3)$ 
8:    $y \leftarrow y + (k_1 + 2k_2 + 2k_3 + k_4)/6$ 
9:    $x \leftarrow x + h$ 
10: end while
11: Return  $y$ 

```

□ Code:-

a) Euler's Method:

#include <stdio.h>

double f(double x, double y) { return x+y; }

int main() {

double x0, y0, h, xn, x, y;

printf("Enter x0, y0, h, xn: ");

scanf("%lf %lf %lf %lf", &x0, &y0, &h, &xn);

x = x0; y = y0;

while(x < xn) {

y = y + h*f(x,y);

x += h;

}

printf("Approximate y(%lf) = %lf\n", xn, y);

return 0;

}

b) Runge-Kutta 2nd & 4th Order:

#include <stdio.h>

double f(double x, double y) { return x+y; }

```
int main() {
    double x0, y0, h, xn, x, y, k1, k2, k3, k4;
    printf("Enter x0, y0, h, xn: ");
    scanf("%lf %lf %lf %lf", &x0, &y0, &h, &xn);

    x = x0; y = y0;
    while(x < xn) {
        // RK4
        k1 = h*f(x,y);
        k2 = h*f(x+h/2, y+k1/2);
        k3 = h*f(x+h/2, y+k2/2);
        k4 = h*f(x+h, y+k3);
        y = y + (k1+2*k2+2*k3+k4)/6;
        x += h;
    }
    printf("RK4 Approximate y(%lf) = %lf\n", xn, y);
    return 0;
}
```

❑ Output:-

```
Enter x0, y0, h, xn:1
2
3
4
Approximate y(4.000000) = 11.000000
```

Figure 7.1

```
Enter x0, y0, h, xn:1
2
3
4
RK4 Approximate y(4.000000) = 60.500000
```

Figure 7.2

❑ Conclusion:-

Problem Statement-8:-

Develop a Python program to calculate the acceleration of a car that starts from rest and reaches a velocity of 10 m/s in 20 seconds, using the formula: $(v_{\text{final}} - v_{\text{initial}})/\text{time}$.

❑ Objective:- The objective of this assignment is to develop a Python program that calculates the acceleration of a car using the formula $(v_{\text{final}} - v_{\text{initial}})/\text{time}$, where the car starts from rest and reaches a velocity of 10 m/s in 20 s.

❑ Algorithm:-

(a) Linear Fit

Algorithm 10 Linear Curve Fit $y = a + bx$

- 1: Compute sums: $\sum x, \sum y, \sum xy, \sum x^2$
 - 2: Solve normal equations:
 - 3: $an + b \sum x = \sum y$
 - 4: $a \sum x + b \sum x^2 = \sum xy$
 - 5: Return a, b
-

(b) Exponential Fit

Algorithm 11 Exponential Fit $y = ae^{bx}$

- 1: Transform to $\ln y = \ln a + bx$
 - 2: Compute $\sum x, \sum \ln y, \sum x \ln y, \sum x^2$
 - 3: Solve normal equations for $\ln a, b$
 - 4: Return $a = e^{\ln a}, b$
-

(c) Polynomial Fit (Quadratic)

Algorithm 12 Quadratic Fit $y = a + bx + cx^2$

- 1: Compute $\sum x, \sum x^2, \sum x^3, \sum x^4, \sum y, \sum xy, \sum x^2y$
 - 2: Solve normal equations:
 - 3: $an + b \sum x + c \sum x^2 = \sum y$
 - 4: $a \sum x + b \sum x^2 + c \sum x^3 = \sum xy$
 - 5: $a \sum x^2 + b \sum x^3 + c \sum x^4 = \sum x^2y$
 - 6: Return a, b, c
-

❑ Code:-

a) Linear Fit $y = a + bx$:

```
#include <stdio.h>
```

```
int main() {
    int n, i;
    double x[20], y[20], sumx=0, sumy=0, sumxy=0, sumx2=0, a, b;
    printf("Enter number of data points: ");
    scanf("%d", &n);
```



```

for(i=0;i<n;i++) {
    scanf("%lf %lf", &x[i], &y[i]);
    sumx+=x[i]; sumy+=y[i];
    sumxy+=x[i]*y[i]; sumx2+=x[i]*x[i];
}
b = (n*sumxy - sumx*sumy)/(n*sumx2 - sumx*sumx);
a = (sumy - b*sumx)/n;
printf("Best fit line: y = %.4f + %.4fx\n", a, b);
return 0;
}

```

b) Exponential Fit → Model:

```

#include <stdio.h>
#include <math.h>

int main() {
    int n, i;
    double x[20], y[20], sumx=0, sumy=0, sumxy=0, sumx2=0;
    double a, b;

    printf("Enter number of data points: ");
    scanf("%d", &n);
    printf("Enter x and y values:\n");
    for(i=0;i<n;i++) {
        scanf("%lf %lf", &x[i], &y[i]);
        sumx += x[i];
        sumy += log(y[i]);    // Taking log of y
        sumxy += x[i]*log(y[i]);
        sumx2 += x[i]*x[i];
    }

    b = (n*sumxy - sumx*sumy) / (n*sumx2 - sumx*sumx);
    a = exp((sumy - b*sumx)/n);

    printf("Best fit exponential curve:
    y = %.4f * e^(%.4f x)\n",

```

```

    a, b);
    return 0;
}

```

c) Polynomial Fit (Quadratic Example) → Model:

```
#include <stdio.h>
```

```

int main() {
    int n, i;
    double x[20], y[20];
    double sumx=0, sumx2=0, sumx3=0, sumx4=0;
    double sumy=0, sumxy=0, sumx2y=0;
    double a, b, c, D, D1, D2, D3;

    printf("Enter number of data points: ");
    scanf("%d", &n);
    printf("Enter x and y values:\n");
    for(i=0;i<n;i++) {
        scanf("%lf %lf", &x[i], &y[i]);
        sumx += x[i];
        sumx2 += x[i]*x[i];
        sumx3 += x[i]*x[i]*x[i];
        sumx4 += x[i]*x[i]*x[i]*x[i];
        sumy += y[i];
        sumxy += x[i]*y[i];
        sumx2y += x[i]*x[i]*y[i];
    }

```

```
// Normal equations in matrix form:
```

```

// | n      sumx    sumx2 |   | a |   | sumy    |
// | sumx    sumx2   sumx3 | * | b | = | sumxy   |
// | sumx2   sumx3   sumx4 |   | c |   | sumx2y  |

```

```

D  = n*(sumx2*sumx4 - sumx3*sumx3)
    - sumx*(sumx*sum4 - sumx2*sumx3)
    + sumx2*(sumx*sumx3 - sumx2*sumx2);

```

```

D1 = sumy*(sumx2*sumx4 - sumx3*sumx3)
    - sumx*(sumxy*sumx4 - sumx3*sumx2y)
    + sumx2*(sumxy*sumx3 - sumx2*sumx2y);

D2 = n*(sumxy*sumx4 - sumx3*sumx2y)
    - sumy*(sumx*sumx4 - sumx2*sumx3)
    + sumx2*(sumx*sumx2y - sumxy*sumx2);

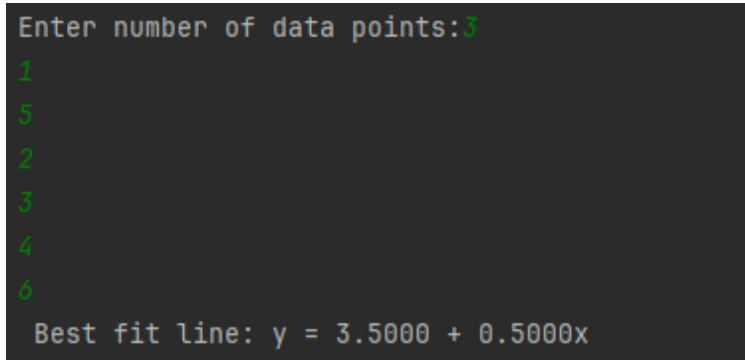
D3 = n*(sumx2*sumx2y - sumxy*sumx3)
    - sumx*(sumx*sumx2y - sumxy*sumx2)
    + sumy*(sumx*sumx3 - sumx2*sumx2);

a = D1/D;
b = D2/D;
c = D3/D;

printf("Best fit quadratic curve:
y = %.4f + %.4fx + %.4fx^2\n", a, b, c);
return 0;
}

```

□ Output:-



```

Enter number of data points:3
1
5
2
3
4
6
Best fit line: y = 3.5000 + 0.5000x

```

Figure 8.1

```
Enter number of data points:3
Enter x and y values:
1
2
2
5
3
7
Best fit exponential curve: y = 1.1775 * e^(0.6264 x)
```

Figure 8.2

```
Output
Enter number of data points: 5
Enter x and y values:
1 2
2 5
3 10
4 17
5 26
Best fit quadratic curve: y = 1.0000 + 0.0000x + 1.0000x^2

=== Code Execution Successful ===
```

Figure 8.3

❑ Conclusion:-

Problem Statement-9:-

Write a program to display Pascal's triangle.

- ❑ **Objective:-** The objective of this assignment is to develop a Python program that generates and displays Pascal's Triangle up to a specified number of rows. This task helps students understand nested loops, conditional logic, and iterative computation. It reinforces the concept of binomial coefficients, where each element in the triangle is derived from the sum of the two elements directly above it. Additionally, the assignment enhances logical thinking, pattern recognition, and structured program design in Python by translating a mathematical pattern into an algorithmic solution.

❑ Algorithm:-

Algorithm 13 Gauss Elimination Method

```

1: Form augmented matrix  $[A|b]$ 
2: for  $k = 1$  to  $n - 1$  do
3:   for  $i = k + 1$  to  $n$  do
4:      $m = a_{ik}/a_{kk}$ 
5:     for  $j = k$  to  $n + 1$  do
6:        $a_{ij} = a_{ij} - m \cdot a_{kj}$ 
7:     end for
8:   end for
9: end for
10: Back substitution:
11:  $x_n = a_{n,n+1}/a_{nn}$ 
12: for  $i = n - 1$  down to  $1$  do
13:    $x_i = (a_{i,n+1} - \sum_{j=i+1}^n a_{ij}x_j)/a_{ii}$ 
14: end for
15: Return solution vector  $x$ 

```

❑ Code:-

```

#include <stdio.h>
#define N 3

int main() {
    float a[N][N+1], x[N];
    int i,j,k;
    float ratio;

    printf("Enter augmented matrix (%dx%d):\n", N, N+1);
    for(i=0;i<N;i++)
        for(j=0;j<=N;j++)
            scanf("%f",&a[i][j]);

    for(i=0;i<N-1;i++){
        for(j=i+1;j<N;j++){
            ratio=a[j][i]/a[i][i];
            for(k=0;k<=N;k++) a[j][k]-=ratio*a[i][k];
        }
    }

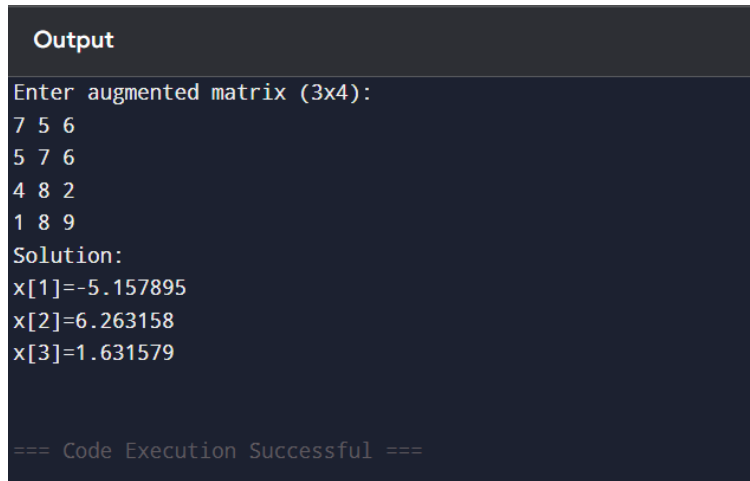
    x[N-1]=a[N-1][N]/a[N-1][N-1];

```

```
for(i=N-2;i>=0;i--){
    x[i]=a[i][N];
    for(j=i+1;j<N;j++) x[i]-=a[i][j]*x[j];
    x[i]/=a[i][i];
}

printf("Solution:\n");
for(i=0;i<N;i++) printf("x[%d]=%f\n",i+1,x[i]);
return 0;
}
```

❑ Output:-



```
Output
Enter augmented matrix (3x4):
7 5 6
5 7 6
4 8 2
1 8 9
Solution:
x[1]=-5.157895
x[2]=6.263158
x[3]=1.631579

=== Code Execution Successful ===
```

Figure 9

❑ Conclusion:-

Acknowledgment

I would like to express my deepest sense of gratitude to my respected teacher, **Mr. Amitava Sardar**, for his valuable guidance, constant encouragement, and continuous support throughout the completion of this Computational Mathematics practical work. His clear and systematic teaching, patient explanations of difficult concepts, and insightful suggestions have not only enhanced my understanding of the subject but have also inspired me to approach problems with greater clarity and confidence. Without his encouragement and dedicated guidance, this work would not have been possible.

I am also sincerely grateful to the **Department of Computer Science** of my college for providing me with the facilities, resources, and academic environment that made this work possible. The availability of proper infrastructure, learning materials, and an atmosphere of academic curiosity has played a vital role in shaping my knowledge and practical skills in computational methods.

I would like to take this opportunity to extend my heartfelt thanks to my **classmates and friends**, who have always been supportive and cooperative throughout the preparation of this record. Their discussions, feedback, and encouragement have helped me remain motivated and complete this work successfully. Finally, I wish to thank everyone who directly or indirectly contributed to the successful completion of this practical work.

Thank you,

[sign. of the mentor]